

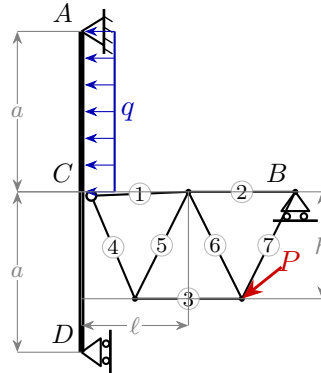
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VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

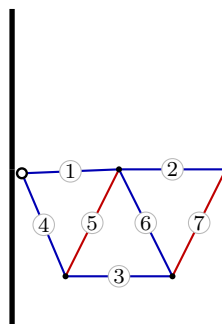
A : $R_x = 14$, $R_y = -3$ D : $R_x = 8$, $R_y = 0$ B : $R_x = 0$, $R_y = 11$

C : $H_x = -10$, $H_y = 3$

3. Truss member forces

Member	Force S [kN]	Type
1	-8.5	compression
2	-5.5	compression
3	-3	compression
4	-3.35	compression
5	3.35	tension
6	-3.35	compression
7	12.3	tension

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams